In [ ]:
```
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```

a. Write a csv file with headers: Student Name, CGPA, Certifications_Count, Internship_Exp_yrs b. Add 4 rows of data c. Print the location of the file. [Hint: absolute_path = os.path.abspath("students.csv")] d. Write the rows one by one e. Write the same data as Key Value Pairs using DictWriter and save in students_dict.csv f. Write another csv file demonstrating the use of header g. Change the mode from w to a and append three more records to the first file h. Read the rows from the first csv file and print them all

In [48]:
```
#question -1

import csv


#a. Write a csv file with headers: Student Name, CGPA, Certifications_Count, Internship_Exp_yrs
#b. Add 4 rows of data

dict={
    "studentname":["Soumya","Santhosh","Neethi","Bhuvana"],"cgpa":[9,9.2,9.3,9.4],"cert_counts":[2,3,4,2],"internsh
}

print(dict)

lst=list(dict)

new_data=["Deep",9.2,4,1.3]

res=lst.append(new_data)
print(lst)

#c. Print the location of the file. [Hint: absolute_path = os.path.abspath("students.csv")]

with.open(dict,
```

```
{'studentname': ['Soumya', 'Santhosh', 'Neethi', 'Bhuvana'], 'cgpa': [9, 9.2, 9.3, 9.4], 'cert_counts': [2, 3, 4,
2], 'internship_yrs': [1, 1.5, 1, 3]}
['studentname', 'cgpa', 'cert_counts', 'internship_yrs', ['Deep', 9.2, 4, 1.3]]
```

Question -4 :

Using Pandas a. Load a student details dictionary to a dataframe b. Use to_json() to save as a json file [hint: use orient] c. Read from the json file and print all the contents d. Inspect the data using head(), info(), describe() methods e. Access the second column and print its contents f. Apply filter on a column and print its contents g. Add a new column and load the values h. Demonstrate updating a value i. Write the updated JSON (dataframe) file

4. Using Pandas a. Load a student details dictionary to a dataframe b. Use to_json() to save as a json file [hint: use orient] c. Read from the json file and print all the contents d. Inspect the data using head(), info(), describe() methods e. Access the second column and print its contents f. Apply filter on a column and print its contents g. Add a new column and load the values h. Demonstrate updating a value i. Write the updated JSON (dataframe) file

In [54]:
```
#4a.Load a student details dictionary to a dataframe

dict={
    "Student_name":["soumya","Santhosh","Deepu","dharani"],
    "cgpa":[9.2,9.3,9.4,9.5],
    "cert_counts":[1,2,3,4],
    "intern_exp":[2,3,1,1.5]
}

df=pd.DataFrame(dict)
print(df)
```

```
  Student_name  cgpa  cert_counts  intern_exp
0       soumya   9.2            1         2.0
1     Santhosh   9.3            2         3.0
2        Deepu   9.4            3         1.0
3      dharani   9.5            4         1.5
```

In [55]:
```
#4d. Inspect the data using head(), info(), describe() methods

df.head()

df.tail()
```

```
df.info()

df.describe()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 4 entries, 0 to 3
Data columns (total 4 columns):
 #   Column        Non-Null Count  Dtype  
---  ------        --------------  -----  
 0   Student_name  4 non-null      str    
 1   cgpa          4 non-null      float64
 2   cert_counts   4 non-null      int64  
 3   intern_exp    4 non-null      float64
dtypes: float64(2), int64(1), str(1)
memory usage: 286.0 bytes
```

Out[55]:

	cgpa	cert_counts	intern_exp
count	4.000000	4.000000	4.000000
mean	9.350000	2.500000	1.875000
std	0.129099	1.290994	0.853913
min	9.200000	1.000000	1.000000
25%	9.275000	1.750000	1.375000
50%	9.350000	2.500000	1.750000
75%	9.425000	3.250000	2.250000
max	9.500000	4.000000	3.000000

In [61]:

```
if df[cgpa]>9:
    print(cgpa)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[61], line 1
----> 1 if df[cgpa]>9:
      2     print(cgpa)

NameError: name 'cgpa' is not defined
```

Question-2. Using Pandas

a. Load student details to a Dataframe b. Use to_csv() to write the the contents from a data frame to csv file c. Read from this csv print and print all contents using read_csv() d. Use the concatenation method to append few more rows to the csv file e. Read the updated file

In [64]:
```
#2a: Load student details to a Dataframe

data={
    "Student_name":["soumya","Santhosh","Deepu","dharani"],
    "cgpa":[9.2,9.3,9.4,9.5],
    "cert_counts":[1,2,3,4],
    "intern_exp":[2,3,1,1.5]
}

df=pd.DataFrame(data)
print(df)
```

```
  Student_name  cgpa  cert_counts  intern_exp
0       soumya   9.2            1         2.0
1     Santhosh   9.3            2         3.0
2        Deepu   9.4            3         1.0
3      dharani   9.5            4         1.5
```

3. Using json a. Create a dictionary of student details as in Question 1 b. Save the dictionary as a JSON file [Hint: use json.dump()] c. Iterate through the JSON file and print the contents d. Read a dictionary as a string and convert that into a Python Dictionary object using json.load() e. Print the contents of the dictionary

In [66]:
```
#3a:Create a dictionary of student details as in Question 1

data={
    "Student_name":["soumya","Santhosh","Deepu","dharani"],
    "cgpa":[9.2,9.3,9.4,9.5],
    "cert_counts":[1,2,3,4],
    "intern_exp":[2,3,1,1.5]
}

df=pd.DataFrame(data)
print(df)
```

```
  Student_name  cgpa  cert_counts  intern_exp
0       soumya   9.2            1         2.0
1     Santhosh   9.3            2         3.0
2        Deepu   9.4            3         1.0
3      dharani   9.5            4         1.5
```

In [ ]:
```
#3b. Save the dictionary as a JSON file [Hint: use json.dump()]

with.open(df,
```

In [ ]:
```

```